

Ambroise Odonnat

PhD student in ML — defense in March 2027

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🐙 [ambroiseodt](https://github.com/ambroiseodt)

🎓 [Google Scholar](https://scholar.google.com/citations?user=ambroiseodt)

Summary

I study how transformers learn and generalize, notably under distribution shift, from task-specific models to large language models and vision foundation models. My work combines theory and large-scale experiments, covering supervised training, pretraining, in-context learning, frozen representations, finetuning, and tool use. I also contributed to test-time scaling in LLMs and to self-improving agents that co-evolve with their evaluators. Next, I want to help build and improve frontier models, notably their reasoning and self-improvement capabilities.

Expertise Science of transformers (representations, training dynamics, theory); distribution shift; label-free evaluation.
Publications ICML (3), NeurIPS (2), ICLR (1), AISTATS (1), ICASSP (1), TMLR (2); 5 as first author.
Impact Oral at ICML 2024 (top 1.5% of submissions); SAMformer deployed in production.

Research Experience

- 2024–present **Huawei Noah's Ark Lab, Research Scientist Assistant, CIFRE PhD program**, Paris.
Advisors: Prof. Romain Tavenard, Prof. Laetitia Chapel, Dr. Ievgen Redko.
◦ Derived generalization bounds for LLMs, developed unsupervised accuracy estimators, analyzed transformer representations for finetuning and probing, and studied tool use in LLMs; experiments on models up to 12B parameters.
◦ **First author of 8 papers**: 3 main conferences ([NeurIPS](#), [ICML](#), [ICASSP](#)); 2 workshops ([ICLR](#), [ICML](#)); 3 preprints.
◦ Contributed to [recursive self-improvement](#) of agents built on frontier models (evaluation on IMO proof grading).
◦ **Leadership**: co-supervised a research intern on test-time scaling in LLMs (paper accepted at [ICML](#)).
- 2023–2024 **Huawei Noah's Ark Lab, Research Scientist**, Paris.
Advisor: Dr. Ievgen Redko.
◦ Co-developed SAMformer, a transformer model for forecasting ([ICML Oral](#)) and a label-free accuracy estimator ([TMLR](#)).
◦ **Internal impact**: SAMformer deployed by business units to forecast battery charge cycles.
- 2023 **Huawei Noah's Ark Lab, Research Intern (6 months)**, Paris.
Advisors: Dr. Ievgen Redko, Dr. Vasilii Feofanov.
◦ Developed a method for robust self-training under sample selection bias ([AISTATS](#)).
- 2022 **Polytechnique Montréal, Research Intern (6 months)**, Montréal.
Advisor: Prof. Julien Cohen-Adad.
◦ Developed a DL pipeline to detect epileptic spikes in EEG ([QBIN conference 2022](#), Best Flash Talk Award).

Education

- 2024–present **PhD in Computer Science**, Univ. Rennes 2, Inria, IRISA & Huawei Noah's Ark Lab.
Understanding Generalization in Transformers: From Specialized to Foundation Models.
Advisors: Prof. Romain Tavenard, Prof. Laetitia Chapel, Dr. Ievgen Redko.
- 2022–2023 **MSc Mathematics, Vision, Learning (MVA)**, ENS Paris-Saclay.
Summa cum laude, GPA 4.0/4.0. Advisor: Prof. Gabriel Peyré.
- 2019–2023 **Engineering Degree (MSc)**, École des Ponts ParisTech, Institut Polytechnique de Paris.
Summa cum laude, GPA 3.9/4.0.
- 2017–2019 **Preparatory Classes (CPGE)**, Lycée Henri IV, MPSI-MP.

Selected Publications (full list)

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| 2026 | Probing Transformer Representations Layer by Layer, Module by Module
<i>first author</i> — LLMs, vision foundation models, frozen representations — extension of ICLR Workshop | Preprint |
| 2026 | The Red Queen Gödel Machine: Co-Evolving Agents and Their Evaluators
<i>co-author</i> — LLMs, self-improving agents — featured in Cambridge CS news | arXiv |
| 2026 | Post-Training in Time Series Foundation Models: A Unifying Framework
<i>first author*</i> — time series foundation models, post-training | arXiv |
| 2026 | Vision Transformer Finetuning Benefits from Non-Smooth Components
<i>first author</i> — vision foundation models, finetuning | ICML |

2026	Optimal Self-Consistency for Efficient Reasoning with LLMs <i>co-supervisor</i> — LLMs, test-time scaling	ICML
2026	Provable Benefits of In-Tool Learning for Large Language Models <i>first author*</i> — LLMs, tool use, finetuning	ICLR Workshop
2026	A Mechanistic Study of Transformers Training Dynamics <i>first author</i> — transformers on arithmetic tasks, supervised training	ICML Workshop
2025	Easing Optimization Paths: a Circuit Perspective <i>first author*</i> — transformers on arithmetic tasks, supervised training	ICASSP
2024	Large Language Models as Markov Chains <i>first author*</i> — LLMs, pretraining, in-context learning — featured in Forbes	arXiv
2024	MaNo: Exploiting Matrix Norm for Unsupervised Accuracy Estimation <i>first author*</i> — distribution shift	NeurIPS
2024	SAMformer: Unlocking the Potential of Transformers in Time Series Forecasting <i>first author*</i> — supervised training, generalization — integrated in PyTorch Forecasting	ICML Oral
2024	Leveraging Ensemble Diversity for Robust Self-Training under Sample Selection Bias <i>first author</i> — supervised training, distribution shift	AISTATS

*Equal contribution.

Awards & Grants

2023	Merit Scholarship , Chair Saint-Gobain & École des Ponts ParisTech (\$5,800)
2022	Research grant , proposal author, Tanenbaum Open Science Institute & McGill University (\$10,800)
2022	Best Flash Talk Award , QBIN Conference (\$1,000)

Invited Talks (full list)

Sep. 2026	Dauphine — A mechanistic study of transformers training dynamics
Mar. 2026	Mila — Vision transformer finetuning benefits from non-smooth components
Mar. 2026	Imperial College London — Provable benefits of tool use for LLMs
Feb. 2026	Kyutai — Provable benefits of tool use for LLMs
Oct. 2025	Cohere — Provable benefits of tool use for LLMs
Jun. 2025	Cohere Labs — Large language models as Markov chains
Feb. 2025	EPFL — Large language models as Markov chains
Sep. 2024	ENS Ulm — SAMformer: transformers for time series forecasting
Jul. 2024	Criteo AI Lab — SAMformer: transformers for time series forecasting

Community Service & Teaching

2026	Conference Organizer , ELLIS UnConference , European ML gathering, ~2000 participants.
2025	Conference Organizer , NeurIPS@Paris , Sorbonne University, ~350 participants.
2025	Workshop Organizer , NeurIPS 2025 , BERT²: Recent Advances in Time Series Foundation Models .
Reviewing	NeurIPS (2025), ICML (2024–2026), ICLR (2025, 2026), AISTATS (2026), TMLR
Teaching	ML teaching assistant (M2), École des Ponts (2024); maths examiner for HEC-ESSEC entrance exam (2024).

Skills

ML stack	Python, PyTorch, Hugging Face Transformers, cuML, NVIDIA Brev
Open source	Reproducible code for my papers (GitHub); contributor to skada , skada-bench , and benchopt .
Languages	French (native), English (fluent), German (beginner)
Other	Model United Nations in Rome (2015); fencing (10 years), university football & volleyball.

References

Charles Arnal, FAIR at Meta
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